



Assessing spatial variability of nutrients, phytoplankton, and related water quality constituents in the California Sacramento-San Joaquin Delta at the landscape scale: 2020-2021 High resolution mapping surveys

Methods

Broad-area, high-resolution, boat-based water quality mapping surveys of the Sacramento-San Joaquin Delta (Delta) were conducted in June, July, August, and October of 2020 and 2021. Each survey was comprised of four successive days of high-resolution mapping. The regions covered included the north-south axis of the Delta from the Sacramento River north of Freeport to the Harvey Banks export facility located in the southern Delta, including Old and Middle Rivers; the east-west axis of the Delta from the Sacramento River at Rio Vista to the San Joaquin River at Stockton, including Disappointment Slough and the North and South Forks of the Mokelumne River; and the northern and western Delta from the northern reaches of the Cache Slough Complex to the Grizzly, Honker, and Suisun Bays. On each day, the route of the mapping survey was determined by weather, tides and other logistical considerations. Data were collected at high speed to minimize the extent to which tidal currents obscure nutrient and phytoplankton distributions.

Definition of regions and reaches for analysis

Aside from evaluating the Delta as a whole, it is useful to divide the survey data into smaller sub-sets for analysis. We chose two main approaches to do this. First, we defined eight Study Regions based on hydrodynamically distinct zones identified by Jabusch and others (2016). Broadly speaking, the Delta has three tidal transition zones (North Delta, Mokelumne System, South Delta), two strongly-tidally-forced tidal lagoons (Western and Central Delta), two dead-end channel systems (Cache Slough Complex, Suisun Marsh), and the brackish part of the estuary that connects the Delta to San Francisco Bay (Suisun Bay). Thus, the eight regions defined in this study and used in the online visualization tool include: (1) North Delta Tidal

Transition Zone, (2) Mokelumne Tidal Transition Zone, (3) Cache Slough Complex, (4) Central Tidal Zone, (5) South Delta Tidal Transition Zone, (6) San Joaquin Tidal Transition Zone, (7) Western Tidal Zone, and (8) Suisun Bay. A 'Transition Zone' is where transport transitions from being mostly unidirectional, like in rivers, to mostly tidal. These eight regions are similar to regions proposed by Jabusch and others (2016) to evaluate data collection efforts and nutrient status and trends analyses in the Delta; those regions were based on Operational Landscape Units (OLUs) originally developed to represent restoration opportunity areas based on an understanding of ecological function, physical drivers, existing constraints, and elevation gradients, while taking into account hydrologic features, watershed boundaries, and DSM2 modeling requirements (Grenier and Grossinger, 2016). Second, we associated data with the geographic name of each water feature (e.g., Sacramento River, Georgiana Slough, Old River) based on widely recognized naming conventions that were cross referenced with the USGS Geographic Names Information System (GNIS) database (USGS, 2021).

Sampling and analysis

Underway high resolution (HR) water sampling and onboard analyses

Surveys were conducted using the USGS R/V Mary Landsteiner and R/V George R. Aiken on successive days, following the approach described by Fichot and others (2015), Downing and others (2016), Kimmerer and others (2019), and Stumpner and others (2020). In 2020: June 8, 9, 10, and 11; July 27 and 28; August 17, 18, and 19; and October 19, 20, 21, and 22; and in 2021: May 11, 12, 13, and 14; July 26, 27, 28, and 29; and October 4, 5, 6, and 7. For analysis, sample water was continuously pumped onto the boat using two SHURflo Aqua King II pumps at a pressure of 60 psi and rate of 5 GPM (SHURflo Pentair, London, United Kingdom) while underway at speeds up to 30 mph (13 m/s) using a pick-up tube mounted at a fixed depth of approximately 1 m below the surface. Water was routed through a 178 μ m in-line strainer (Cole Parmer; EW-29595-47, Vernon Hills, Illinois) to remove large debris. Sample water was split into separate flow paths as describe below, and each flow path was routed through a pressure-compensated manifold that maintained system pressure at a proscribed level (~20 PSI) irrespective of boat speed (Downing and others, 2016). A 2-stage debubbler was used to remove bubbles can interfere with instrument performance. Flow rates were maintained by metering the discharge line, except for the flow path serving the open split interface, which was metered on the feed line. Flows were continuously monitored using a sight gauge (0.1-1 GPM Flow Meter LZM-15 with adjustable knob) for each flow path.

A manifold delivered unfiltered water to the following flow paths: (1) A flow-through system consisting of a thermosalinograph that recorded temperature, conductance, and salinity (Sea-Bird Scientific SB45 (TSG), Belluve, Washington); fluorometers that measured fluorescence of chlorophyll (fCHL) and fluorescence of dissolved organic matter (fDOM, Sea-Bird Scientific

WETstar (WS), Philomath, Oregon); a beam transmissometer that recorded transmittance (Sea-Bird Scientific model C-Star transmissometer (CStar), Philomath, Oregon); a nitrate (NO₃) analyzer (SUNA V2; Sea-Bird Scientific, Bellevue, Washington). (2) A second path provides water to a multiparameter water quality sonde (YSI EXO2; Xylem Inc. (EXO), Rye Brook, New York) equipped with sensors to measure temperature, specific conductance (SpC), turbidity, pH, dissolved oxygen (DO), fDOM and fCHL. (3) A third path passed water to a fluorometer designed to measure different algal classes (FluoroProbe III (FP); BBE Moldaenke, Schwentinental, Germany; Eijkelkamp, 2014; Beutler and others, 2002).

A second manifold was setup to provide 0.2 micron filtered (Osmonics Memtrex, 25 cm length, 0.2 µm pore size; MNY921EGS; Osmonics, Inc.) water to an open-split interface at atmospheric pressure, which served water to the on-board ammonium (NH₄) analyzer. The NH₄ analyzer was a continuous flow, gas diffusion/conductivity-based (Carlson, 1978) analyzer for ammonium (NH₄) analysis (TL-2800; Timberline Instruments, Boulder, Colorado) that was modified for field operation and continuous data collection by the manufacturer (Thompson and others, 2023). Modifications included installation in a ruggedized housing, addition of an automated line-switching valve, addition of a heating unit to maintain the instrument at a constant above-ambient temperature, and changes to the software. The analyzer was run in continuous mode with frequent periodic introduction of deionized organic-free water (OFW, resistivity >18.2 MΩ/cm), and standard solutions to continuously assess instrument performance and to correct for baseline drift during the day. Full standard curves were run at the beginning and end of each day, and partial curves were run throughout the day.

All instrumentation was cleaned, and calibrations were checked prior to each use following the manufacturer's recommendation or as described above. Data for most instruments were recorded at 1-second frequency on a single data logger (CR6, Campbell Scientific, Logan, Utah) together with a timestamp and boat position obtained from a high-resolution global positioning system receiver (16X-HVS, Garmin, Olathe, Kansas). The FluoroProbe logged data internally and to the host software every 1-2 seconds. The Timberline NH₄ analyzer was connected to a stand-alone computer and collected data through its native software. All data were displayed in real time, so scientists on board could respond when they observed changes relevant to study objectives, or if issues with flow or instrument performance were noted. Data collected in situ are referred to as high resolution (HR) parameters.

Discrete water sampling and laboratory analyses

Discrete water samples (DS) were collected either while underway or—in the case of more extensive sample requirements—while stopped on station. Discrete samples were collected while underway by passing water from the pickup tube through a 0.2-micron filter into a sample bottle. At designated sampling stations (approximately 10 per day) samples for

determination of NH_4 , NO_2 , NO_3 , PO_4 , dissolved organic carbon (DOC), optics (absorbance and fluorescence), and total dissolved nitrogen (TDN) concentrations were collected using this system. Additionally, unfiltered samples were collected approximately from 1 m depth using a submersible pump for analysis of chlorophyll a (Chl a) concentration, pheophytin concentration, phytoplankton enumeration, and picophytoplankton enumeration. Samples for phytoplankton enumeration and picophytoplankton enumeration were fixed immediately with Lugol's iodine solution (~5%) and formaldehyde (~2%), respectively.

Most samples were stored on wet ice in the dark during transit to the laboratory where they were stored at 2°C. Samples for picophytoplankton analysis were stored on dry ice in the dark during transit to the laboratory where picophytoplankton samples were stored at -80°C. Dissolved organic carbon samples were preserved to <2 pH with high-purity sulfuric acid the day following collection and stored at 2°C. Samples for pigment analysis were filtered within 24 hours of collection and immediately frozen (-80°C). Total Chl a samples were filtered through a 0.7 μm nominal pore size glass fiber filter (Advantec MFS, Inc, Dublin, California). Chl a associated with the larger sized cell fraction were filtered through a 5 μm pore size 25 mm polycarbonate filter (General Electric Healthcare, Chicago, Illinois). Frozen Chl a filters and chilled nutrient samples were shipped to the USGS National Water Quality Laboratory (Denver, Colorado). Phytoplankton and picophytoplankton enumeration samples were shipped to BSA Environmental Services, Inc. (Beachwood, Ohio) for analysis.

Concentrations of nutrients were determined colorimetrically using an automated-segmented flow analyzer. Concentrations of nitrogen as nitrite ($\text{NO}_2\text{-N}$) and as nitrate plus nitrite ($\text{NO}_3\text{-N} + \text{NO}_2\text{-N}$) were determined by colorimetric analysis (Fishman, 1993; Patton and Kryskalla, 2011), ammonium as nitrogen ($\text{NH}_4\text{-N}$) was determined by colorimetric analysis after reaction with salicylate-hypochlorite (Fishman, 1993), and total dissolved nitrogen (TDN) was determined by alkaline persulfate digestion (Patton and Kryskalla 2003). Orthophosphate as phosphorous ($\text{PO}_4\text{-P}$; also referred to as soluble reactive phosphorus) was determined by colorimetric analysis after reaction with NH_4 molybdate and reduction with ascorbic acid (Patton and Truitt, 1992).

Chl a and phaeophytin concentrations were determined according to EPA Method 445.0 (Arar and Collins, 1997); 0.7 micron filtered samples were extracted using sonication while 5 micron filtered samples were extracted using mechanical grinding.

Dissolved organic carbon concentrations were measured using a total organic carbon analyzer (TOC-VCSH, Shimadzu Scientific Instruments, Columbia, Maryland) using high-temperature catalytic combustion according to a modified version of USEPA method 415.3 (Potter and Wimsatt, 2012). Absorbance spectra and fluorescence EEMs were simultaneously measured using an Aqualog® equipped with a photodiode and charge-coupled device (CCD) (Horiba

Instruments, NJ, USA) according to the method described by Hansen and others (2018). Briefly, excitation and absorbance spectra were measured using a 150 W xenon lamp, a 5 nm bandpass, and a 1 s integration time at wavelengths of 240–600 nm. Emission spectra were collected with a CCD at approximately 1.64 nm (4 pixel) intervals at wavelengths of 250–600 nm. To reduce ultraviolet light exposure of the sample and limit the effects of photobleaching, excitation and absorbance wavelengths were scanned from low to high energy (i.e., visible to ultraviolet range) during analysis. Peaks, ratios, and slopes reported in the data were calculated as described by Hansen and others (2016).

Phytoplankton enumeration was completed by BSA Environmental Services in accordance with the American Public Health Association (APHA) Standard Method 10200 (2023). This method focuses on counting and identifying microplankton (20–200 microns) and nanoplankton (2 to 20 microns). Briefly, phytoplankton were enumerated to the lowest possible taxonomic level using membrane-filtered slides, as described in McNabb (1960), and counted using a Leica DMLB compound microscope. The magnification used (100X - 1000X) depended upon the size of the dominant taxa and presence of particulates to ensure that enumeration and identification of taxa of various sizes was achieved. A minimum of 400 natural units (colonies, filaments, and unicells) or a minimum of 50 fields were counted from each sample. Counting 400 natural units provides accuracy within 90-percent confidence limits (Lund and others, 1958). In addition, an entire strip of the filter was counted at high magnification (usually 630X) along with one-half of the filter at a lower magnification (usually 400X) to ensure complete species reporting. Cell biovolume of all identified taxa were estimated using formulae for solid geometric shapes that most closely match the cell shape; method by Hillebrand and others (1999). Biovolume calculations were based on measurements of 10 organisms per taxon for each sample, where possible. For taxa with substantial size variation (such as diatoms), size classes were designated arbitrarily to determine average cell size (biovolume). For each taxon, 10 cells were measured from each size class (assuming that enough were available). From Burkholder and Wetzel (1989), the mean biovolume within each size class were used to calculate the total biovolume contributed by the taxon to its representative sample.

Picophytoplankton were enumerated via epifluorescence microscopy as described by Murrell and Lores (2004) using a Leica DM2500 epifluorescence microscope equipped with a Xenon light source as well as multiple excitation and emission filter sets at 1000X total magnification. Phycoerythrin-containing cells (PE) and phycocyanin-containing cells (PC) were counted and measured separately and then summed to obtain total picophytoplankton densities and biovolume for cells sized between 0.2 and 2.0 microns. For each sample, a minimum of 300 cells were counted unless this tally was not reached within 100 fields.

Data processing

Final continuous NO₃ values were obtained by regressing the SUNA instrument response against NO₃ concentrations obtained from laboratory measurements of discrete samples collected through the course of the day (Pellerin and others, 2013). Individual sample results greater than three standard deviations from the regression were judged to be outliers and removed from the regression. Larger differences between SUNA and laboratory NO₃ concentration data are assumed to arise from 1) actual spatial heterogeneity in environmental NO₃ concentrations between water entering the flow through system and that captured in the sample bottle; 2) interferences of turbidity and dissolved organic matter using the UV absorption method for nitrate; or 3) errors in sample handling and laboratory analysis. Excepting these potential sources of error, the field and laboratory NO₃ measurements are typically well correlated with a linear fit. Please refer to the “Data Corrections and Offsets” tab of the Data Dictionary for individual regression equations.

Timberline continuous NH₄ data was processed by first correcting the voltage output of the instrument for baseline drift based on the instrument response during periodic measurements of organic-free deionized water and then using a regression model of the voltage response to a standard concentration. A regression coefficient of $>0.97 R^2$ or better was judged to be acceptable. Additional data processing steps include the removal of “bad data” caused by operator error or aberrations due to electrical interference or bubbles.

All data, except that collected by the Timberline, were processed using the pandas software library (0.220; McKinney, 2010) in Python (3.7.3; Van Rossum and Drake, 2009) to remove periods of compromised data (e.g. flow blockages, bubbles), to apply instrument corrections and unit conversions, and to apply a centered 20 second median filter to the time series. Data from onboard continuous instruments not directly logged to the flow-through data collection system (e.g., the Timberline ammonium analyzer) were merged based on timestamp to the nearest second after estimating the differential travel times between flow paths. These time lags accounted for the additional time it took for filtered sample water to reach the instruments and the time to travel through the instruments to the detectors and for the response to be reported by the software. The time differential arises from lag introduced by filtering and specifics of instrument operation that determine the time it takes for environmental sample water to reach the instrument’s detector and for a response to be recorded.

Dissolved inorganic nitrogen (DIN) concentration was calculated as the sum of NO₃, NO₂ and NH₄ for the continuous data and for the discrete sample data. Continuous DOC and DIN concentrations were developed by regressing onboard fDOM data against laboratory determined DOC and DIN concentrations measured in discrete samples (Snyder and others, 2018). These data are reported as estDOC and estDIN. Please refer to the “Data Corrections and Offsets” tab of the Data Dictionary for individual regression equations and R² values. Data exceeding 3 times the standard deviation of the residuals was judged to be an outlier and not included in the model.

Laboratory reported data for discrete samples from the National Water Quality Information System (NWIS; USGS, 2022) are included for convenience in a discrete data file. Accompanying laboratory reported data are NWIS qualifier and remark codes as they appear in NWIS (U.S. Geological Survey, 2011a, 2011b). For discrete nutrient sample concentrations, in addition to reporting nutrient parameters as they appear in NWIS (e.g., mg-N/L or mg-P/L), we also reported those same parameters converted to µM units. For laboratory reported values that are reported below the method lowest detection limit (LDL), half of the LDL is reported in the associated µM column (Clarke, 1998), and those data were used to bias correct the SUNA NO₃ data (Pellerin and others, 2013) and to generate the spatially aligned datasets. Table 1 summarizes mg/L to µM conversion factors and ½ the LDL values (Clarke, 1998).

Table 1. Summary of conversions factors used to calculate micromolar nutrient concentrations, NWQL method lowest detection limit (LDL), and ½ LDLs used for bias correction and model calculations.

Parameter	Conversion factor	LDL (mg-/L)	LDL (µM)	1/2 LDL µM
NO₃	71.39	0.04	2.86	1.428
NH₄	71.39	0.01	0.71	0.36
NO₂	71.39	0.001	0.071	0.036
TDN	71.39	0.05	3.57	1.785
PO₄	32.28	0.004	0.13	0.065

Spatial Alignment (SA)

Data were spatially aligned to geographic coordinate system [NAD 1983 (2011)] to facilitate comparisons between dates and to minimize the effects of differential data density on the spatial interpolation calculations. The spatial alignment was performed using the GeoPandas software library in python (Jordahl, 2014), which assigned median-filtered data points to polygon shapefiles that are spaced at approximately 150 m intervals. Spatially aligned data were then aggregated by polygon shapefile to obtain summary statistics of all median-filtered data within a polygon.

Interpolated water quality data were created from the spatially aligned data using ESRI's ArcGIS Pro Spline with Barriers tool (Terzopoulos and Witkin, 1988). The raster output was smoothed with the Focal Statistics tool, and interpolated values from the raster output were extracted to the spatially aligned points with the Extract Multi Values to Points tool to create continuous spatially aligned data across the mapping domain. Rasters were converted to vector format using the Contour tool, then clipped with the Clip tool using a Sacramento-San Joaquin Delta polygon obtained from the National Hydrography Dataset (U.S. Geological Survey, 2016). These data are referred to as spatially aligned (SA) and are associated with a specific sampling day, and water quality parameter.

References

- American Public Health Association (APHA), American Water Works Association, Water Environment Federation, Lipps W.C., Braun-Howland E.B., Baxter T.E., eds., 2023, Standard Methods for the Examination of Water and Wastewater, 24th ed, Washington DC: APHA Press.
- Arar, E.J. and Collins, G.B., 1997, Method 445.0: In vitro determination of chlorophyll a and pheophytin a in marine and freshwater algae by fluorescence: Washington, D.C., U. S. Environmental Protection Agency, Office of Research and Development, National Exposure Research Laboratory. [Also available at https://cfpub.epa.gov/si/si_public_record_report.cfm?Lab=NERL&dirEntryId=309417.]
- Beutler, M., Wiltshire, K., Meyer, B., Moldaenke, C., Lürling, C., Meyerhöfer, M., & Dau, H., 2002, A fluorometric method for the differentiation of algal populations in vivo and in situ: Photosynthesis Research, v. 72, p. 39-53, <https://doi.org/10.1023/A:1016026607048>.
- Burkholder, J.B. and Wetzel, R.G., 1989, Microbial colonization on natural and artificial macrophytes in a phosphorus-limited, Hardwater Lake: Journal of Phycology, v. 25, p. 55-65, <https://doi.org/10.1111/j.0022-3646.1989.00055.x>.
- Carlson, R. M., 1978, Automated separation and conductimetric determination of ammonia and dissolved carbon dioxide: Analytical Chemistry, v. 50, p. 1528-1531, <https://doi.org/10.1021/ac50033a035>.

- Clarke, J.U., 1998, Evaluation of censored data methods to allow statistical comparisons among very small samples with below detection limit observations: *Environmental Science & Technology*, v. 32, no. 1, p. 177-183, <https://pubs.acs.org/doi/10.1021/es970521v>.
- Downing, B.D., Bergamaschi, B.A., Kendall, C., Kraus, T.E.C., Dennis, K.J., Carter, J.A. and Von Dessonneck, T.S., 2016, Using continuous underway isotope measurements to map water residence time in hydrodynamically complex tidal environments: *Environmental Science & Technology*, v. 50, no. 24, p. 13387-13396, <https://doi.org/10.1021/acs.est.6b05745>.
- Eijkelpamp, 2014. FluoroProbe User Manual. V. E2, accessed March 15, 2017, at: https://en.eijkelpamp.com/products/sensors-monitoring_uk/fluoroprobe-iii-with-accessories.html.
- Fichot, C.G., Downing, B.D., Bergamaschi, B.A., Windham-Myers, L., Marvin-DiPasquale, M., Thompson, D.R. and Gierach, M.M., 2016. High-resolution remote sensing of water quality in the San Francisco Bay–Delta Estuary: *Environmental Science & Technology*, v. 50, no. 2, p. 573-583, <https://doi.org/10.1021/acs.est.5b03518>.
- Fishman, M.J., 1993, Methods of analysis by the U.S. Geological Survey National Water Quality Laboratory- Determination of inorganic and organic constituents in water and fluvial sediments: U.S. Geological Survey Open-File Report 93-125, 217 p., <https://doi.org/10.3133/ofr93125>.
- Grenier, L., and Grossinger, R., 2016, Tools for landscape-scale restoration planning in the Delta. Presentation to the Delta Independent Science Board, January 16, 2013, 2nd Floor Conference Center, Sacramento, CA.
- Hansen, A.M., Kraus, T.E.C., Pellerin, B. A., Fleck, J. A., Downing, B. D., Bergamaschi, B. A., 2016, Optical properties of dissolved organic Matter (DOM): effects of biological and photolytic degradation: *Limnology and Oceanography*, v. 61, p. 1015-1032, <https://doi.org/10.1002/lno.10270>.
- Hansen, A. M., Fleck, J. A., Kraus, T. E. C., Downing, B. D., Von Dessonneck, T., Bergamaschi, B. A., 2018, Procedures for using the Horiba Scientific Aqualog® fluorometer to measure absorbance and fluorescence from dissolved organic matter: U. S. Geological Survey Open-File Report 2018-1096, 31 p., <https://doi.org/10.3133/ofr20181096>.
- Hillebrand, H., Dürselen, C., Kirschtel, D., Pollinger, U., Zohary, T., 1999, Biovolume calculation for pelagic and benthic microalgae: *Journal of Phycology* v. 35, p. 403-424, <https://doi.org/10.1046/j.1529-8817.1999.3520403.x>.
- Jabusch, T., Bresnahan, P., Trowbridge, P., Novick, E., Wong, A., Salomon, M. and Senn, D., 2016, Summary and evaluation of Delta subregions for nutrient monitoring and assessment: Richmond, Calif. San Francisco Estuary Institute, Contribution No. 789, <https://www.sfei.org/documents/delta-subregions>.
- Jordahl, K., 2014, GeoPandas: Python tools for geographic data, accessed December 2, 2020, at <https://geopandas.org/en/stable/>.
- Kimmerer, W., Wilkerson, F., Downing, B., Dugdale, R., Gross, E.S., Kayfetz, K., Khanna, S., Parker, A.E. and Thompson, J., 2019, Effects of drought and the emergency drought barrier on the ecosystem of the California Delta: *San Francisco Estuary and Watershed Science*, v. 17, no. 3, <https://doi.org/10.15447/sfews.2019v17iss3art2>.

- Lund, J. W. G., Kipling, C., Le Cren, E. D., 1958, The inverted microscope method of estimating algal numbers and the statistical basis of estimations by counting: *Hydrobiologia*, v. 11, p. 143-170. <https://doi.org/10.1007/BF00007865>.
- McKinney, W., 2010, Data structures for statistical computing in python: *Proceedings of the 9th Python in Science Conference*, v. 445, p. 51–56, <https://doi.org/10.25080/Majora-92bf1922-00a>.
- McNabb, C. D., 1960, Enumeration of freshwater phytoplankton concentrated on the membrane filter: *Limnology and Oceanography*, v. 5, p. 57-61, <https://doi.org/10.4319/lo.1960.5.1.0057>.
- Murrell, M.C. and Lores, E.M., 2004, Phytoplankton and zooplankton seasonal dynamics in a subtropical estuary: importance of cyanobacteria: *Journal of Plankton Research*, v. 26, p. 371-382, <https://doi.org/10.1093/plankt/fbh038>.
- Patton, C.J., Kryskalla, J.R., 2003, Methods of analysis by the U.S. Geological Survey National Water Quality Laboratory-Evaluation of alkaline persulfate digestion as an alternative to Kjeldahl digestion for determination of total and dissolved nitrogen and phosphorus in water: U.S. Geological Survey Water-Resources Investigations Report 2003-4174, 33 p., <https://doi.org/10.3133/wri034174>.
- Patton, C.J. and Kryskalla, J.R., 2011, Colorimetric determination of nitrate plus nitrite in water by enzymatic reduction, automated discrete analyzer methods: U.S. Geological Survey Techniques and Methods 5B-8, 34 p., <https://doi.org/10.3133/tm5B8>.
- Patton, C.J. and Truitt, E.P., 1992, Methods of analysis by the U.S. Geological Survey National Water Quality Laboratory-Determination of the total phosphorus by a Kjeldahl digestion method and an automated colorimetric finish that includes dialysis: US Geological Survey Open-File Report 92-146, 39 p., <https://doi.org/10.3133/ofr92146>.
- Pellerin, B.A., Bergamaschi, B.A., Downing, B.D., Saraceno, J.F., Garrett, J.A., and Olsen, L.D., 2013, Optical techniques for the determination of nitrate in environmental waters: guidelines for instrument selection, operation, deployment, maintenance, quality assurance, and data reporting: U.S. Geological Survey Techniques and Methods 1–D5, 37 p., <https://doi.org/10.3133/tm1D5>.
- Potter, B. and Wimsatt, J., 2012, USEPA method 415.3: quantifying TOC, DOC, and SUVA: *Journal American Water Works Association*, v. 104, 6, p. E358-E369, <https://doi.org/10.5942/jawwa.2012.104.0086>.
- Snyder, L., Potter, J.D., and McDowell, W.H., 2018. An evaluation of nitrate, fDOM, and turbidity sensors in New Hampshire streams: *Water Resources Research*, v. 54, no. 3, p. 2466-2479, <https://doi.org/10.1002/2017WR020678>.
- Stumpner, E. B., Bergamaschi, B. A., Kraus, T. E. C., Parker, A. E., Wilkerson, F. P., Downing, B. D., Dugdale, R. C. Murrell, M. C., Carpenter, K. D., Orlando, J. L., and Kendall, C., 2020, Spatial variability of phytoplankton in a shallow tidal freshwater system reveals complex controls on abundance and community structure: *Science of The Total Environment*, v. 700, no. 134392, 17 p., <https://doi.org/10.1016/j.scitotenv.2019.134392>.
- Terzopoulos, D. and Witkin, A., 1988, Physically based models with rigid and deformable components: *IEEE Computer Graphics and Applications*, v. 8, no. 6, p. 41-51, <https://doi.org/10.1109/38.20317>.

Thompson, L., Mussen, T.D., Cook, M., Nordin, J., Noss, J., Bigler, U., Ramamoorthy, S., Berg, G.M., Driscoll, S., Herrmann, C., Kimmerer, W., Ignoffo, T., Kraus, T.E.C., Fackrell, J., Bergamaschi, B., Guerin, M., and Rachiele, R., 2023, Sacramento River nutrient change study. A report to the Delta Regional Monitoring Program, State Water Contractors, and U.S. Bureau of Reclamation, https://deltarmp.org/Documents/Sacramento_River_Nutrient_Change_Study.pdf.

U.S. Geological Survey, 2011a, Value Qualifier Codes: U.S. Geological Survey National Water Information System: Help System, https://help.waterdata.usgs.gov/code/val_qual_cd_query.

U.S. Geological Survey, 2011b, Remark Codes: U.S. Geological Survey, National Water Information System: Help System, https://help.waterdata.usgs.gov/remark_cd.

U.S. Geological Survey, 2016, National Hydrography Dataset (NHD): U. S. Geological Survey, National Geospatial Technical Operations Center (NGTOC), accessed December 14, 2018, at <https://www.usgs.gov/national-hydrography/national-hydrography-dataset>.

U.S. Geological Survey, 2021, Geographic names information system (GNIS) website: U.S. Geological Survey, accessed June 19, 2023, at <https://www.usgs.gov/tools/geographic-names-information-system-gnis>.

U.S. Geological Survey, 2022, USGS water data for the Nation: U.S. Geological Survey, National Water Information System database, <http://dx.doi.org/10.5066/F7P55KJN>.

Van Rossum, G., and Drake, F. L., 2009, Python 3 Reference Manual. Scotts Valley, CA: CreateSpace, accessed March 15, 2017.